

Power-Up Reference

AdventureMultiplayer | All Power-Ups & Traps

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1. Inventory System

Every player carries 3 power-up slots. Slots fill when a player walks into a NetworkedPowerUpBox. Tapping a slot on the HUD calls UseSlotServerRpc. All effect logic runs on the SERVER and results are pushed to clients via targeted ClientRpc calls. Stats ScriptableObjects are cloned per player so multiple players can run the same power-up type simultaneously without interfering.

2. Self-Use Power-Ups

2.1 Speed Boost

Effect: 1.4x top speed, 1.2x acceleration and air-acceleration for 5 seconds.

Targeting: Self only. No shield or invincibility interaction.

How it works

- BeginSpeedBoostClientRpc multiplies all stat entries on the owner client.
- After 5 s the multipliers are divided back to restore original values.
- Re-using while active cancels the previous timer and resets the 5 s window.

2.2 Shield

Effect: Absorbs the next incoming hostile power-up, then is consumed.

Max duration: 30 seconds (auto-expires if never hit).

Blocks (consumed): Rocket, StunBolt, Freeze, Swap.

Does NOT block: Banana Peel and Decoy Box (both intentionally bypass shield).

If Invincibility is active when a hit arrives, the shield is not consumed.

2.3 Invincibility

Effect: Blocks all hostile power-ups for 2 seconds without being consumed by hits.

Blocks (kept): Rocket, StunBolt, Freeze, Swap, Banana Peel.

Does NOT block: Decoy Box (intentionally bypasses invincibility).

3. Offensive Power-Ups

3.1 Rocket

Targeting

- Always targets the player currently in 1st place.
- If the caster IS in 1st place, the power-up has NO effect (slot consumed, nothing fires).

Projectile Behaviour

- Homing rocket: 20 u/s travel speed, 180 deg/s steering toward the target.
- Auto-despawns after 10 seconds if it never hits.
- Kinematic Rigidbody (MovePosition/MoveRotation), replicated via NetworkTransform.

On Hit

- Shield on target: shield absorbed (consumed), rocket despawns - no damage.
- Invincibility on target: rocket blocked (invincibility kept), rocket despawns - no damage.
- Otherwise: KILL SHOT - deals full MaxHealth damage. Target dies and respawns.
 - Health set to 0 on server; all clients HUD updated via ForceSetHealthClientRpc.
 - TriggerOwnerDeathClientRpc sent only to the victim's owner client.
 - player.Die() fires playerEvents.OnDie, which starts NetworkRespawner.
 - After 2 s player respawns at last checkpoint with arc offset to avoid stacking.
 - SyncRespawnHealthServerRpc restores full health on server and all clients.

3.2 StunBolt

Targeting

- Targets the player directly ahead in race position.
- Fallback: if caster is in 1st, targets the player directly behind instead.
- No other player found: power-up has no effect.

Shield / Invincibility

- Shield: absorbs hit (consumed), no stun applied.
- Invincibility: blocks hit (not consumed), no stun applied.

Stun Effect (3 seconds)

- Zeros lateralVelocity and verticalVelocity every LateUpdate.
- Sets animator.speed = 0 - character pose freezes, movement animation stops.
- Clears all AI input flags (move direction, jump, dash, spin, stomp).
- On expiry: animator.speed restored to 1, full control returns.

3.3 Freeze

Targeting

- Stuns ALL players behind the caster simultaneously.
- Each target checked independently for shield / invincibility.

Shield / Invincibility (per target)

- Shield: absorbs (consumed per player). Invincibility: blocks (not consumed per player).

Stun Effect (default 10 seconds - Inspector configurable)

- Identical to StunBolt: zeros velocity, freezes animation, clears AI input.

3.4 Swap

Targeting

- Swaps position with the player directly ahead in race ranking.
- No effect if the caster is already in 1st place.

Shield / Invincibility on target

- Shield: absorbs (consumed), swap cancelled entirely.
- Invincibility: blocks (not consumed), swap cancelled entirely.

Effect

- TeleportClientRpc moves each player to the other's world position.
- Both players' velocities (lateral and vertical) are zeroed on arrival.
- RaceManager.SwapCheckpoints swaps checkpoint indices so ranking reflects new positions.

4. Trap Power-Ups

4.1 Banana Peel

Placement

- Dropped 1.2 units behind the caster at ground level.
- Persists for 15 seconds or until one player triggers it (single-use trap).
- The player who placed the peel is IMMUNE to their own trap.

On Contact with another player

- Invincibility on victim: slip blocked (not consumed). Trap stays active.
- Shield on victim: does NOT protect - no shield check in BananaPeel.
- Otherwise: trap despawns, ApplySlipClientRpc sent to victim's owner client.

Slip Effect (default 2 seconds - Inspector configurable)

- player.lateralVelocity set to forward x 10 u/s so minSpeedToSlide threshold is met.
- player.states.Change<CrouchPlayerState>() - shrinks collider, enables ground-snap.
- animator.SetBool("Is Sliding", true) - animator moves to the Slide state.
 - Saphy / MixamoPlayer: plays Saphy|RailGrind animation clip.
 - Lily: plays Lily|Grind animation clip.
- Every Update while grounded: lateralVelocity locked to forward x 10 u/s (constant glide).
- Airborne: upward velocity suppressed - player cannot jump out of the slide.
- On expiry: Is Sliding = false, state reverts naturally.

Setup: run Tools > Adventure Multiplayer > Setup Slide Animation once to wire the Slide state and "Is Sliding" parameter into all animator controllers.

4.2 Decoy Box

Placement

- Dropped 3 units ahead + 0.5 units up (lands in path in front of the caster).
- Persists for 20 seconds or until one player triggers it (single-use trap).
- The player who placed it is IMMUNE to their own trap.

On Contact with another player

- INTENTIONALLY bypasses BOTH Shield and Invincibility - no protection against this trap.
- Trap despawns, ApplyStunClientRpc sent to victim: 2 second StunEffect.
- Same stun as StunBolt/Freeze: zeros velocity, freezes animation, clears AI input.

5. Rocket Kill - Respawn Flow

The complete server-to-client call chain when a Rocket lands a kill shot:

```
RocketProjectile.OnTriggerEnter (server)
+- NetworkedHealth.ApplyDamageFromPowerUp(maxHealth)
+- Health.Value = 0
+- ForceSetHealthClientRpc(0)      --> ALL clients: health.Set(0), HUD update
+- TriggerOwnerDeathClientRpc      --> OWNER client only:
  +- player.Die()                  --> playerEvents.OnDie fires
    | +- NetworkRespawner.OnPlayerDied()
    |   +- RespawnAfterDelay() (2 second UniTask wait)
    |     +- ApplyRespawn()
    |       +- player.SetRespawn(checkpoint + arc offset)
    |       +- player.Respawn()      health restored locally
    |       +- SyncRespawnHealthServerRpc()
    |       +- m_health.Set(max) on server
    |       +- ForceSetHealthClientRpc(max) --> all clients
+- player.states.Change<DiePlayerState>() freeze pose during the 2s wait
```

IMPORTANT: DiePlayerState.OnEnter is empty in PLAYER TWO - calling Change<DiePlayerState> does NOT fire

playerEvents.OnDie. Only player.Die() fires OnDie. Always call player.Die() before changing state, never after.

6. Shield & Invincibility Interaction Matrix

Power-Up	Shield	Invincibility
Rocket	Absorbs (shield consumed)	Blocks (kept)
StunBolt	Absorbs (shield consumed)	Blocks (kept)
Freeze	Absorbs (shield consumed)	Blocks (kept)
Swap	Absorbs (shield consumed)	Blocks (kept)
Banana Peel	No check - hits through	Blocks (kept)
Decoy Box	No check - hits through	No check - hits through

7. Inspector-Configurable Values

Field	Component	Default	Notes
respawnDelay	NetworkRespawner	2 s	Delay before respawn after death
freezeDuration	PlayerPowerUpInventory	10 s	Freeze stun duration
bananaSlipDuration	PlayerPowerUpInventory	2 s	Banana peel slip duration
slideSpeed	SlipEffect	10 u/s	Speed during banana slide
ShieldMaxDur (const)	PlayerPowerUpInventory	30 s	Shield max duration before expiry
InvincDur (const)	PlayerPowerUpInventory	2 s	Invincibility duration
StunBoltDur (const)	PlayerPowerUpInventory	3 s	StunBolt stun duration
DecoyStunDur (const)	PlayerPowerUpInventory	2 s	DecoyBox stun duration
moveSpeed	RocketProjectile	20 u/s	Rocket travel speed
turnSpeed	RocketProjectile	180 d/s	Rocket steering rate
maxLifetime	RocketProjectile	10 s	Rocket auto-despawn if no hit
lifetime (BananaPeel)	BananaPeel	15 s	Peel auto-despawn lifetime
lifetime (DecoyBox)	DecoyBoxPickup	20 s	Decoy auto-despawn lifetime